

Compare Results

Old File:

USB_PD_R3_2 V1.0 2023-10_Part21.pdf

3 pages (350 KB)

15/04/2024 10:52:44

versus

New File:

USB_PD_R3_2 V1.1 2024-10_Appendix A CRC.pdf

1 page (83 KB)

28/10/2024 11:27:52

Total Changes

5

Text only comparison

Content

- 2 Replacements
- 1 Insertion
- 2 Deletions

Styling and Annotations

- 0 Styling
- 0 Annotations

Go to First Change (page 1)

A CRC calculation

A.1 C code example

```
//
// USB PD CRC Demo Code.
//
#include <stdio.h>
int crc;
//-----
void crcBits(int x, int len) {
    const int poly = 0x04C11DB6; //spec 04C1 1DB7h
    int newbit, newword, rl_crc;
    for(int i=0; i<len; i++) {
        newbit = ((crc>>31) ^ ((x>>i)&1)) & 1;
        if(newbit) newword=poly; else newword=0;
        rl_crc = (crc<<1) | newbit;
        crc = rl_crc ^ newword;
        printf("%2d newbit=%d, x>>i=0x%x, crc=0x%x\n", i, newbit, (x>>i), crc);
    }
}

int crcWrap(int c){
    int ret = 0;
    int j, bit;
    c = ~c;
    printf("~crc=0x%x\n", c);
    for(int i=0; i<32; i++) {
        j = 31-i;
        bit = (c>>i) & 1;
        ret |= bit<<j;
    }
    return ret;
}
//-----
int main(){
    int txCrc=0, rxCrc=0, residue=0, data;
    printf("using packet data 0x%x\n", data=0x0101);
    crc = 0xffffffff;
    crcBits(data, 16);
    txCrc = crcWrap(crc);
    printf("crc=0x%x, txCrc=0x%x\n", crc, txCrc);
    printf("received packet after decode= 0x%x, 0x%x\n", data, txCrc);
    crc = 0xffffffff;
    crcBits(data, 16);
    rxCrc = crcWrap(crc);
    printf("Crc of the received packet data is (of course) =0x%x\n", rxCrc);
    printf("continue by running the transmit crc through the crc\n");
    crcBits(rxCrc, 32);
    printf("Now the crc residue is 0x%x\n", crc);
    printf("should be 0xc704dd7b\n");
}
```